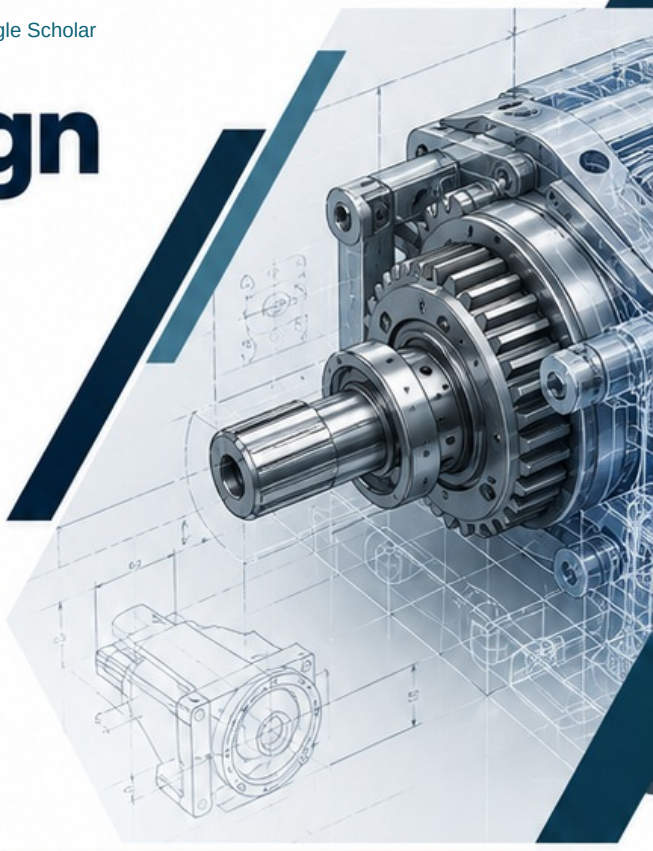
CONTACT Rome, Italy | +39 351 857 1266 | golmoradi17@gmail.com | LinkedIn | Google Scholar

Mechanical Design Portfolio



Milad Golmoradi

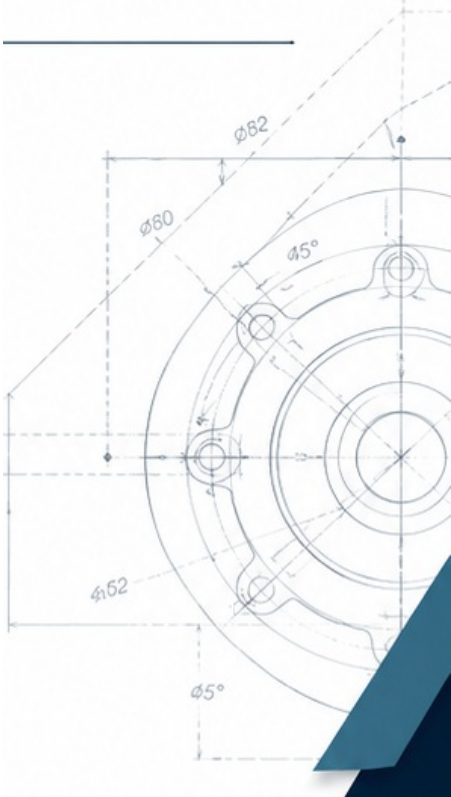
Mechanical Design & Mechatronics Engineer

Mechanical engineer with a Master's degree in Mechanical Engineering and current MSc studies in Mechatronics Engineering at the University of Rome Tor Vergata. Focused on CAD modeling, technical drawings, reverse engineering, dynamic simulation, and interdisciplinary engineering projects.

Core Skills

- CATIA V5
- SolidWorks
- Autodesk Inventor
- AutoCAD
- Fusion 360
- MATLAB / Simulink
- Technical Drawings
- Reverse Engineering
- Dynamic Simulation
- Condition Monitoring

Selected Work in This Portfolio

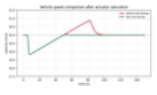


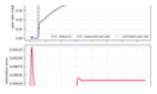
01 
Nail Punching Robot
End-Effector Design & Dynamic Simulation

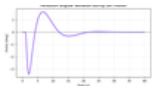
02 
uPVC Mullion Cutting Machine
Reverse Engineering & 3D CAD Modeling

03 
DC-DC Buck Converter
Modeling & Closed-Loop Control

04 
Moss Detection on Walls
Classical Vision & Deep Learning

05 
Cruise & Adaptive Cruise Control
Speed, Distance & Anti-Windup Control

06 
BMW 730 Active Steering
Vehicle Dynamics & State Estimation

07 
Inverted Pendulum Control
LQR Stabilization & Balance Tracking

08 
Research & Publications
Fault Diagnosis & Condition Monitoring

Nail Punching Robot

End-Effector Design & Dynamic Simulation

Academic group project | MSc in Mechatronics | University of Rome Tor Vergata | 2024



Overview

This project explored the design and analysis of a robotic system for automated nail punching. The work covered robot selection, kinematics, workspace, path planning, sensing, and dynamic analysis.

My Contribution

- Designed a custom aluminum holder in CATIA V5 to mount the nail gun to the robot end effector.
- Considered mechanical integration with the last robot joint and the effect of nail-gun recoil.
- Supported dynamic simulation in Autodesk Inventor and MATLAB-based inverse dynamics analysis.
- Contributed within a university group project.

Tools

CATIA V5

Autodesk Inventor

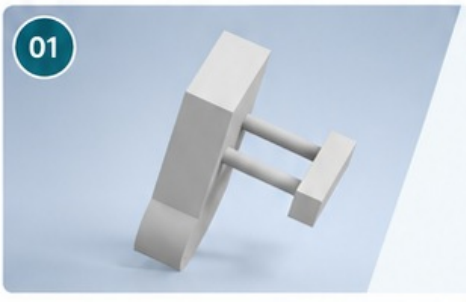
MATLAB

Key Takeaways

Mechanical integration, custom CAD design, dynamic simulation, and robotics workflow understanding.

Project Highlights

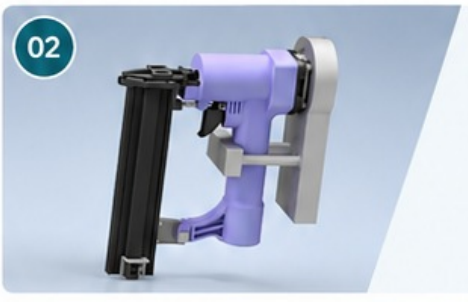
01



Custom holder design

Aluminum tool holder designed in CATIA V5 to mount the nail gun to the robot end effector.

02



Integrated end-effector

Nail gun mounted on the customized holder and connected to the last robot joint.

03



Inventor simulation

Dynamic simulation in Autodesk Inventor to evaluate motion and joint behavior.

04



Motion / dynamics

End-effector trace and dynamic results (velocity, acceleration and torque) for each joint.

uPVC Mullion Cutting Machine



Reverse Engineering & 3D CAD Modeling

Bachelor's project | CATIA-based modeling

This project focused on studying an industrial mullion cutting machine and recreating its structure and mechanism through CAD-based reverse engineering.

What I Did

- Studied the machine structure, cutting mechanism, and major subassemblies.
- Built 3D component and assembly models in CATIA.
- Produced application-environment views and rendered visualizations.
- Created a digital representation useful for understanding machine structure and function.

Tools

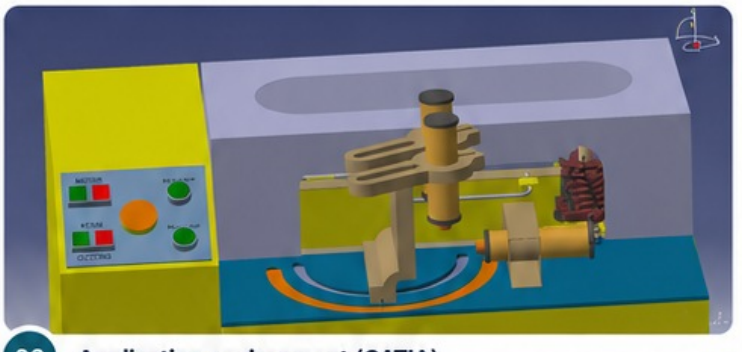
- CATIA
- Reverse Engineering
- Assembly Modeling

Project Focus

Industrial machine understanding, CAD reconstruction, and visual communication of structure and mechanism.

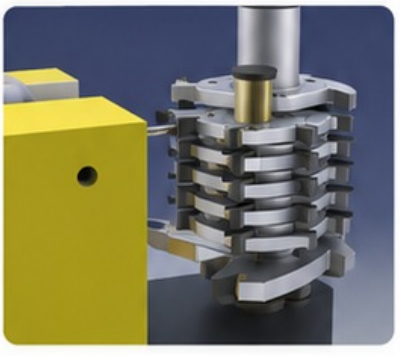


01 Original machine

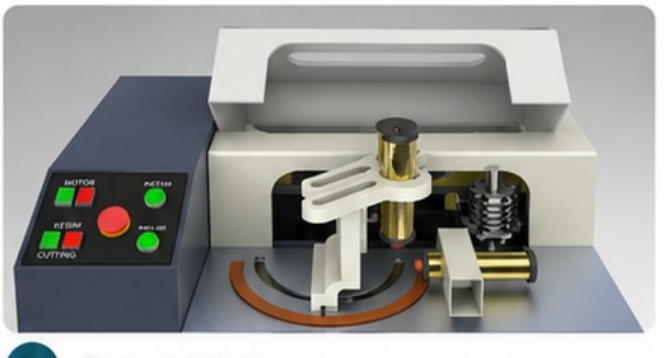


02 Application environment (CATIA)





03 3D model views (CATIA)



04 Final renderings

Selected Mechatronics Projects

01 | DC–DC Buck Converter

Modeling & Closed-Loop Control

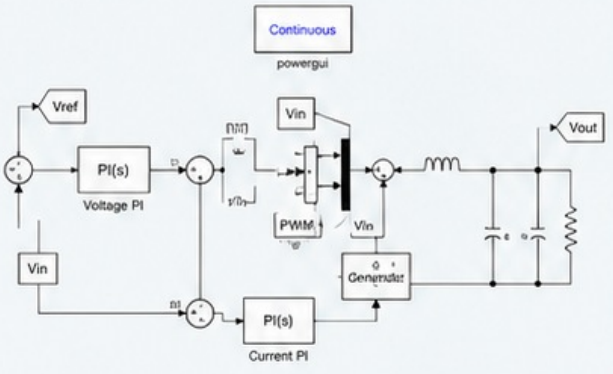
Individual project | MATLAB / Simulink | 2025

Designed and simulated a buck converter under different load and input-voltage conditions using cascaded PI current and voltage control loops.

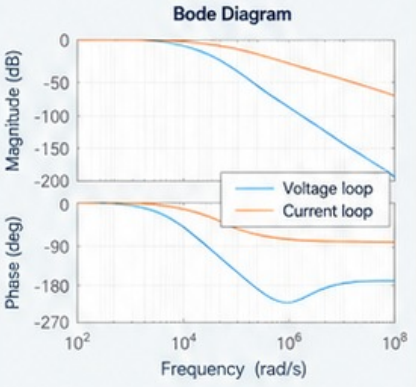
- Built the Simulink model of the power stage and control system.
- Analyzed CCM and DCM behavior under varying conditions.
- Evaluated step responses, Bode plots, and voltage regulation performance.

Key Specifications

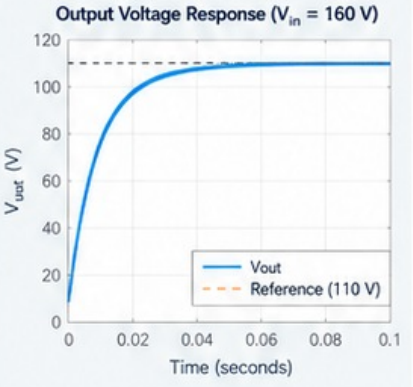
V_{out}	= 110 V
V_{in}	= 140 – 180 V
f_s	= 40 kHz



Simulink model of the buck converter power stage and control system.



Open-loop Bode diagram of voltage and current control loops.



Step response of output voltage at V_in = 160 V (full load).

02 | Moss Detection on Walls

Classical Vision & Deep Learning

Group project | Computer Vision | 2025

Compared a hybrid image-processing approach with a DeepLabv3+ segmentation model for moss detection on wall surfaces.

- Tested a direct method based on edge detection and active contour refinement.
- Evaluated a deep-learning approach using DeepLabv3+ with ResNet-50.
- Compared performance using FM, CR, and CM metrics.

Classical Image Processing (Direct Approach)



Deep Learning (DeepLabv3+)



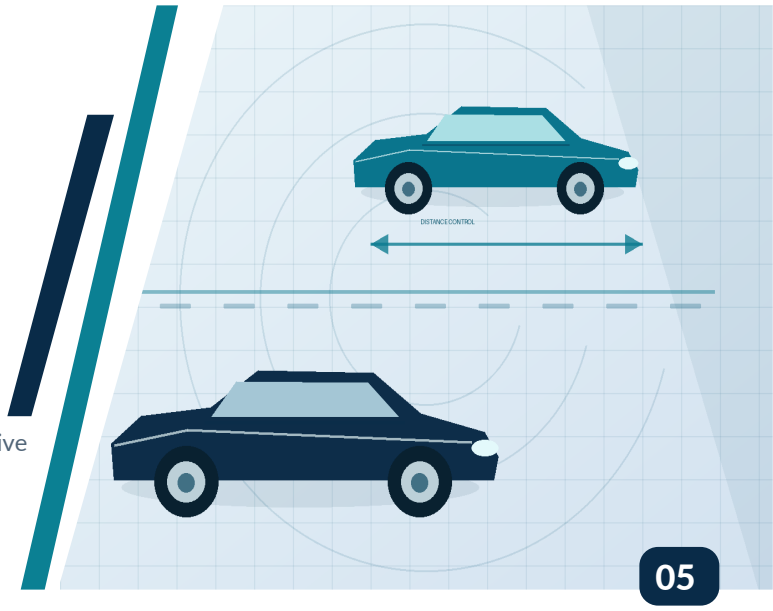
These projects highlight breadth across control, simulation, and data-driven engineering workflows.

Cruise & Adaptive Control

Vehicle Speed & Following-Distance Control

SELECTED ENGINEERING PROJECT

Dynamic modeling and simulation of vehicle speed regulation and adaptive following-distance controllers, including actuator saturation and disturbances.



05

My Contribution

- Built and analyzed the vehicle longitudinal model under road-slope and mass variations.
- Evaluated PI speed regulation with anti-windup correction for throttle saturation.
- Compared PID and observer-based controllers for adaptive following-distance tracking.

Tools

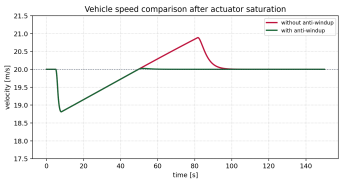
MATLAB | SIMULINK | PI / PID

KEY TAKEAWAY

Anti-windup reduced speed overshoot following saturation; both approaches regulated distance in the simulated leader-speed scenario.

Project Highlights

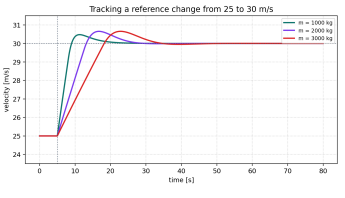
01



Anti-windup response

Measured the improvement in speed recovery after actuator saturation.

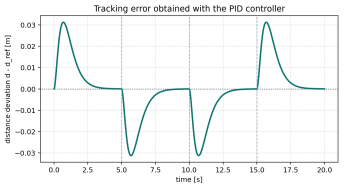
02



Mass sensitivity

Compared tracking response for different vehicle masses.

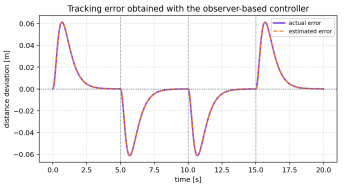
03



PID distance control

Evaluated following-distance error under changes in leader speed.

04



Observer-based control

Compared output-feedback tracking without direct differentiation.

BMW 730
Active Steering

Lateral Dynamics & State Estimation

SELECTED ENGINEERING PROJECT

Investigated the stability and controllability of a vehicle lateral model, then designed active-steering feedback and state observers.



My Contribution

- Analyzed vehicle stability, reachability and observability across operating speeds.
- Investigated the effect of tire-road friction on damping and lateral behavior.
- Designed yaw-rate feedback and compared Luenberger and Wiener/Kalman observers.

Tools

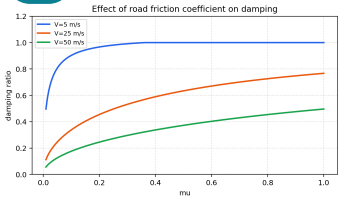
MATLAB | STATE SPACE | KALMAN

KEY TAKEAWAY

State estimation supported yaw-rate feedback in the simulation; lower road friction significantly changed lateral damping.

Project Highlights

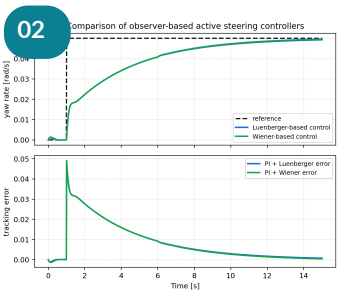
01



Friction effect

Studied how tire-road friction influences the lateral system damping.

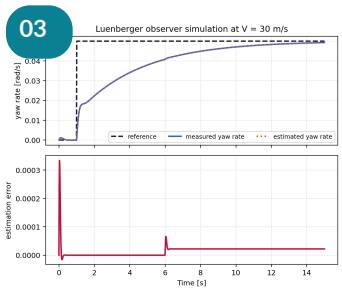
02



Steering feedback

Analyzed active-steering response in the vehicle model.

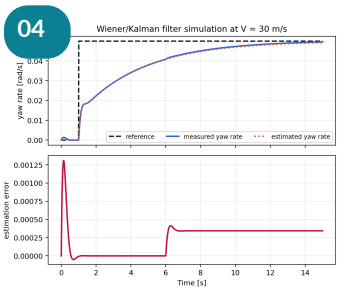
03



Luenberger observer

Examined tracking and estimation error with state feedback.

04



Wiener/Kalman filter

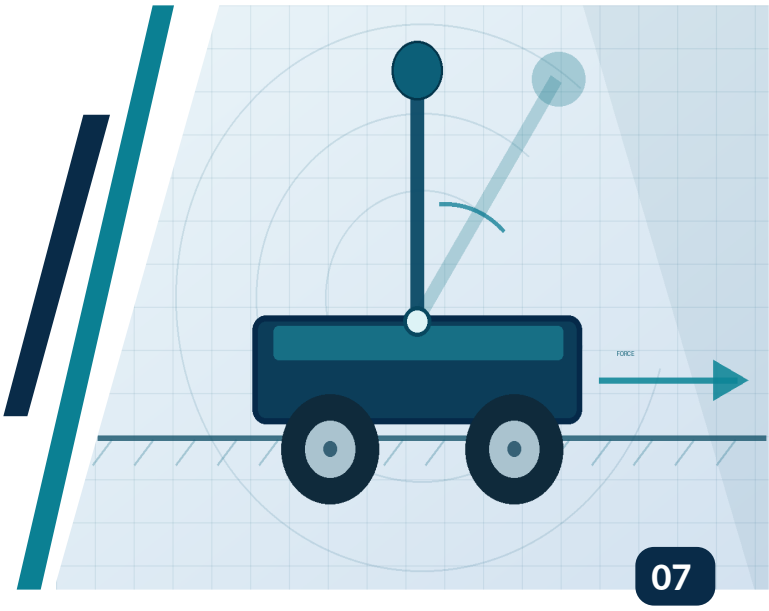
Compared noise-aware state estimation performance.

Inverted Pendulum Control

LQR Stabilization & Position Tracking

SELECTED ENGINEERING PROJECT

Designed and simulated state-space controllers for an unstable cart-pendulum system, balancing upright stabilization and reference tracking.



My Contribution

- Modeled the nonlinear cart-pendulum dynamics and linearized the upright operating point.
- Compared pole-placement control with optimal LQR state feedback.
- Added integral action for nonlinear-model position tracking with an actuator-force limit.

Tools

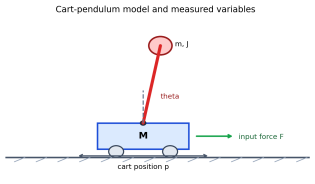
MATLAB | SIMULINK | LQR

KEY TAKEAWAY

The simulated cart tracked a 5-to-8 m position change with zero steady-state error while the pendulum returned upright.

Project Highlights

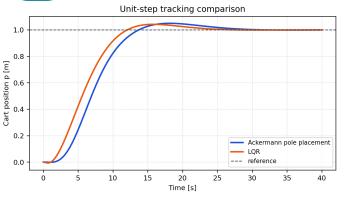
01



Dynamic model

Cart-pendulum equations and simulated mechanical system.

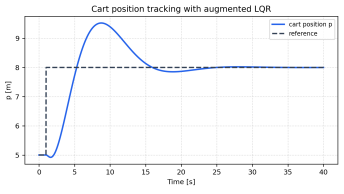
02



LQR vs pole placement

Compared closed-loop cart responses for both feedback strategies.

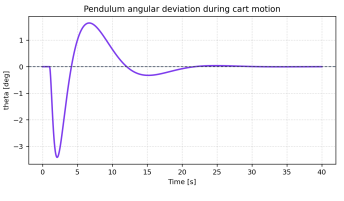
03



Position tracking

Tracked the 5 m to 8 m step using augmented control.

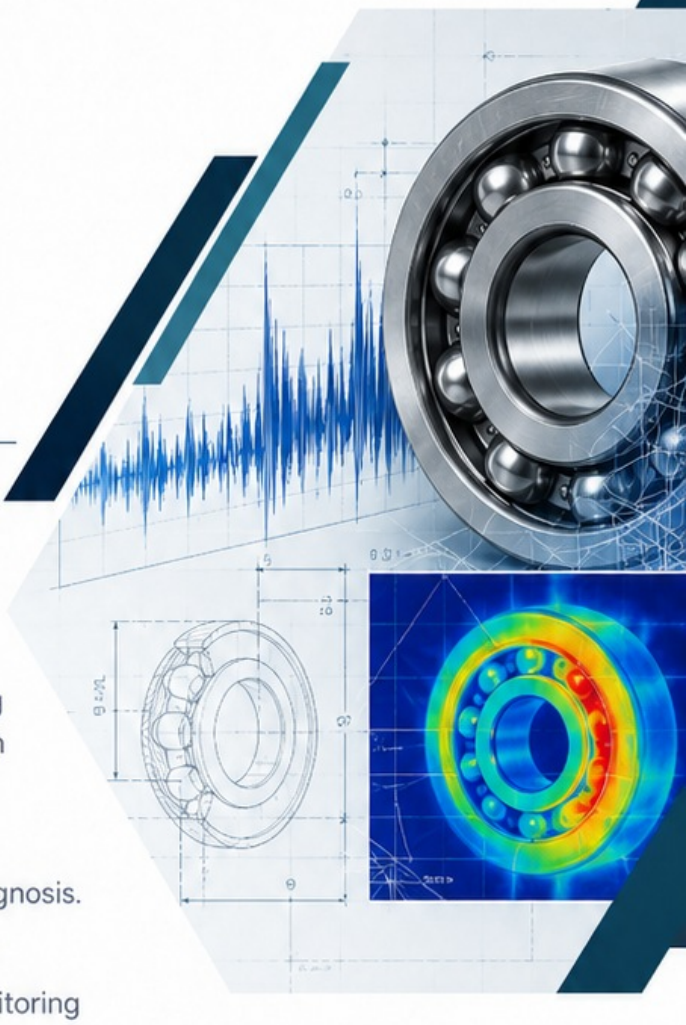
04



Pendulum balance

Angle returns toward the upright equilibrium after motion.

Research & Publications



Thesis Project

Rolling Bearing Fault Prediction Using Data Fusion

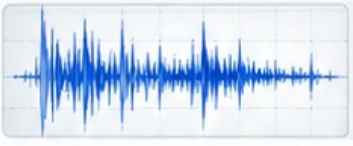
Master's thesis | 2018

Investigated fault prediction of rolling bearings by combining vibration analysis and thermal imaging. The work focused on multimodal feature extraction and SVM-based classification for condition monitoring.

- Combined vibration and thermal information for fault diagnosis.
- Explored feature extraction and classification workflows.
- Demonstrated strong interest in machine condition monitoring and diagnostics.

Selected Publications

01



Compressor Fault Diagnosis Based on SVM and GA DOI

Vibroengineering Procedia, 2017.

02



Fault Diagnosis of Compressor Based on Decision Tree and Fuzzy Inference System DOI

Vibroengineering Procedia, 2017.

Research Interests

- Fault diagnosis
- Vibration analysis
- Condition monitoring
- Signal processing
- Data-driven classification

This portfolio combines mechanical design, simulation, mechatronics, and research experience.